



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2013
Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
NUMBER

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H2 BIOLOGY

9648/02

Paper 2 Core Paper

18 September 2013

2 hours

Additional Materials: Answer papers

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination,

1. Fasten your answer papers to section B securely together.
2. Hand in the following separately:
 - Section A (Part I)
 - Section A (Part II)
 - Section B

The number of marks is given in brackets [] at the end of each question or part question.

For examiner's Use	
Section A	
1	/ 10
2	/ 12
3	/ 12
4	/ 14
5	/ 9
6	/ 13
7	/ 10
Section B	
8 / 9	/ 20
Total	/100

This paper consists of **21** printed pages.

[Turn over]

Section A (Part I)
Answer **all** the questions in this section.

For
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QUESTION 1

Heavy metal ions are waste products of many industrial processes and at times may leak into the ocean and cause environmental pollution. The toxic effects of heavy metals on organisms stems from their effect on metabolically-important enzymes, such as lactate dehydrogenase. Lactate dehydrogenase catalyses the conversion of pyruvate into lactate.

Fig. 1.1 shows the effect of varying concentrations of mercury ions on the rate of reaction catalysed by lactate dehydrogenase isolated from *Oreochromis mossambicus*, a species of fish.

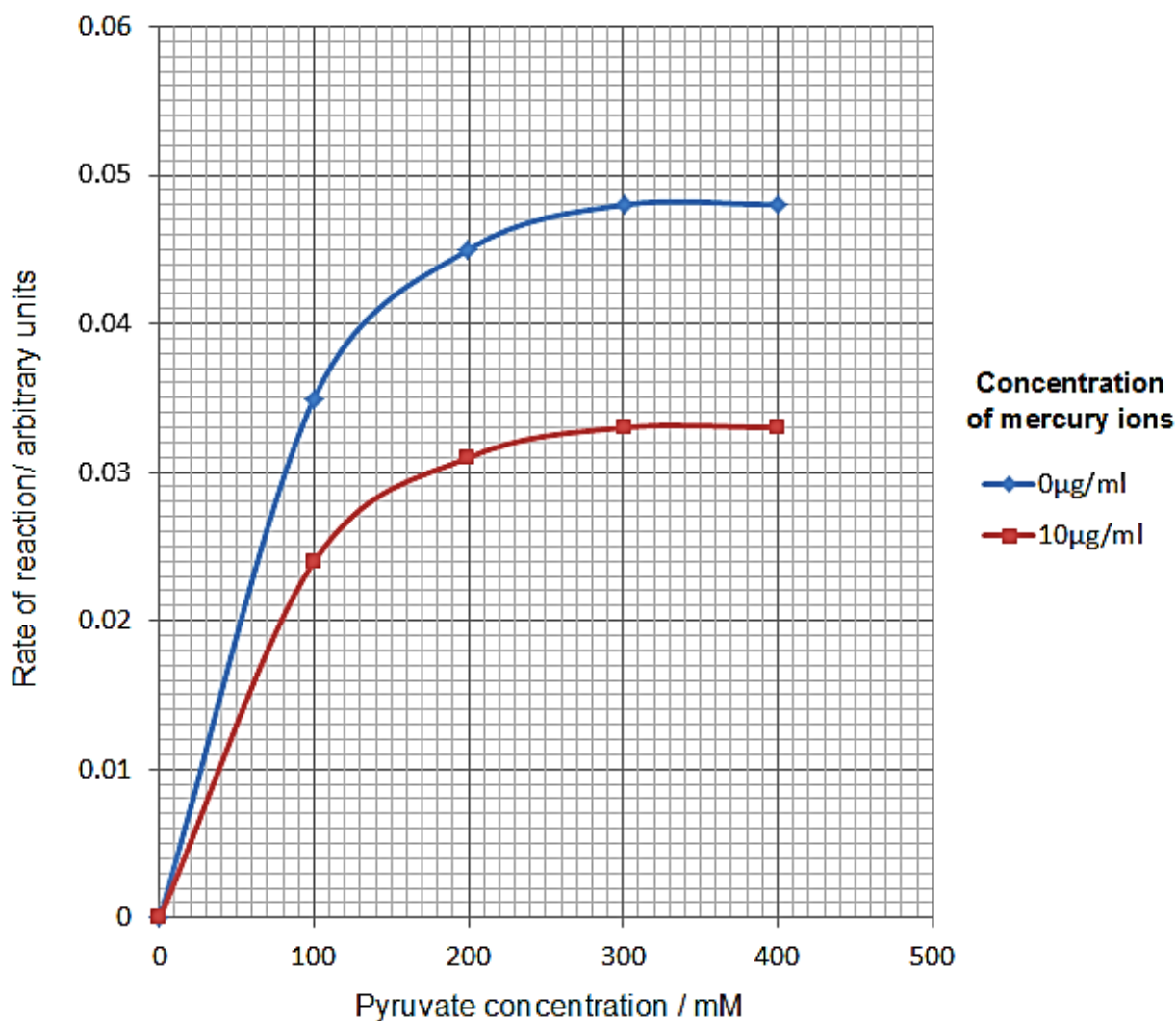


Fig. 1.1

- (a)** Describe and explain the effect of mercury ions on the rate of reaction catalysed by lactate dehydrogenase. [4]

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- (b)** Explain how mercury ions may affect the survival of the fish when oxygen concentrations are low. [3]

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Human lysozyme is an important enzyme which is part of the human body's defence against some bacteria. It is found in abundance in tears, saliva and mucous. The main action of lysozyme is to catalyse the hydrolysis of β -(1,4) glycosidic bonds between N-acetylmuramate (NAM) and N-acetylglucosamine (NAG) sugar residues in the peptidoglycan cell walls of bacteria. With the weakened cell wall, the bacterial cells are more likely to undergo lysis.

Fig. 1.2 shows the structure of lysozyme.

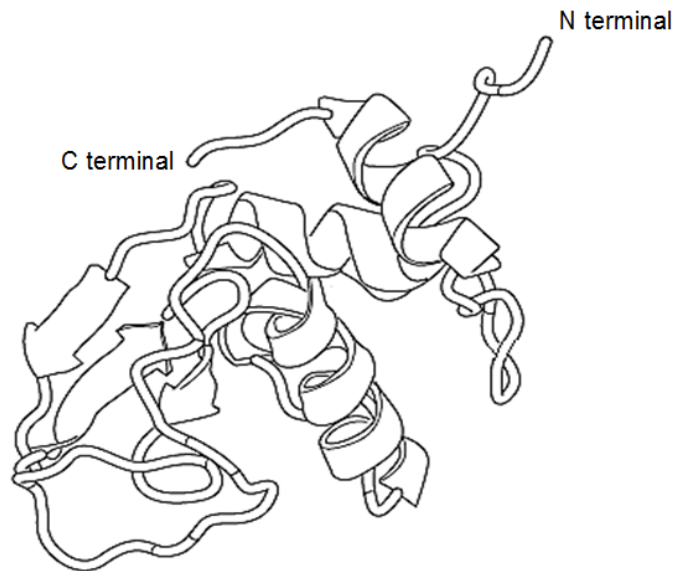


Fig. 1.2

(c) With reference to Fig. 1.2 and with your knowledge of haemoglobin, state two structural differences between lysozyme and haemoglobin. [2]

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2.
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(d) Suggest why some bacteria are more resistant to digestion by lysozymes. [1]

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[Total: 10]

QUESTION 2

Fig. 2.1 illustrates the Central Dogma, which describes the flow of genetic information from DNA to RNA to proteins.

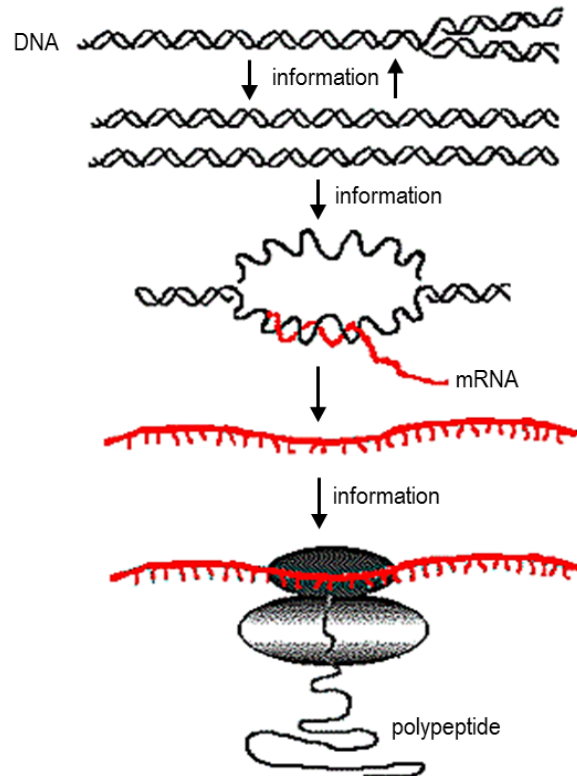


Fig. 2.1

(a) Explain, with an example, how the Central Dogma may not always hold true. [2]

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(b) Describe how the information carried on mRNA allows the synthesis of a complete polypeptide chain. [3]

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A ribonucleoprotein is a protein that is associated with RNA. The ribosome is an example of a ribonucleoprotein. Prokaryotes have 70S ribosomes while eukaryotes have 80S ribosomes, each consisting of a small and large subunit. In prokaryotes, the small subunit is made up of a 16S rRNA associated with 21 proteins. Fig. 2.2 shows the structure of a 16S rRNA.

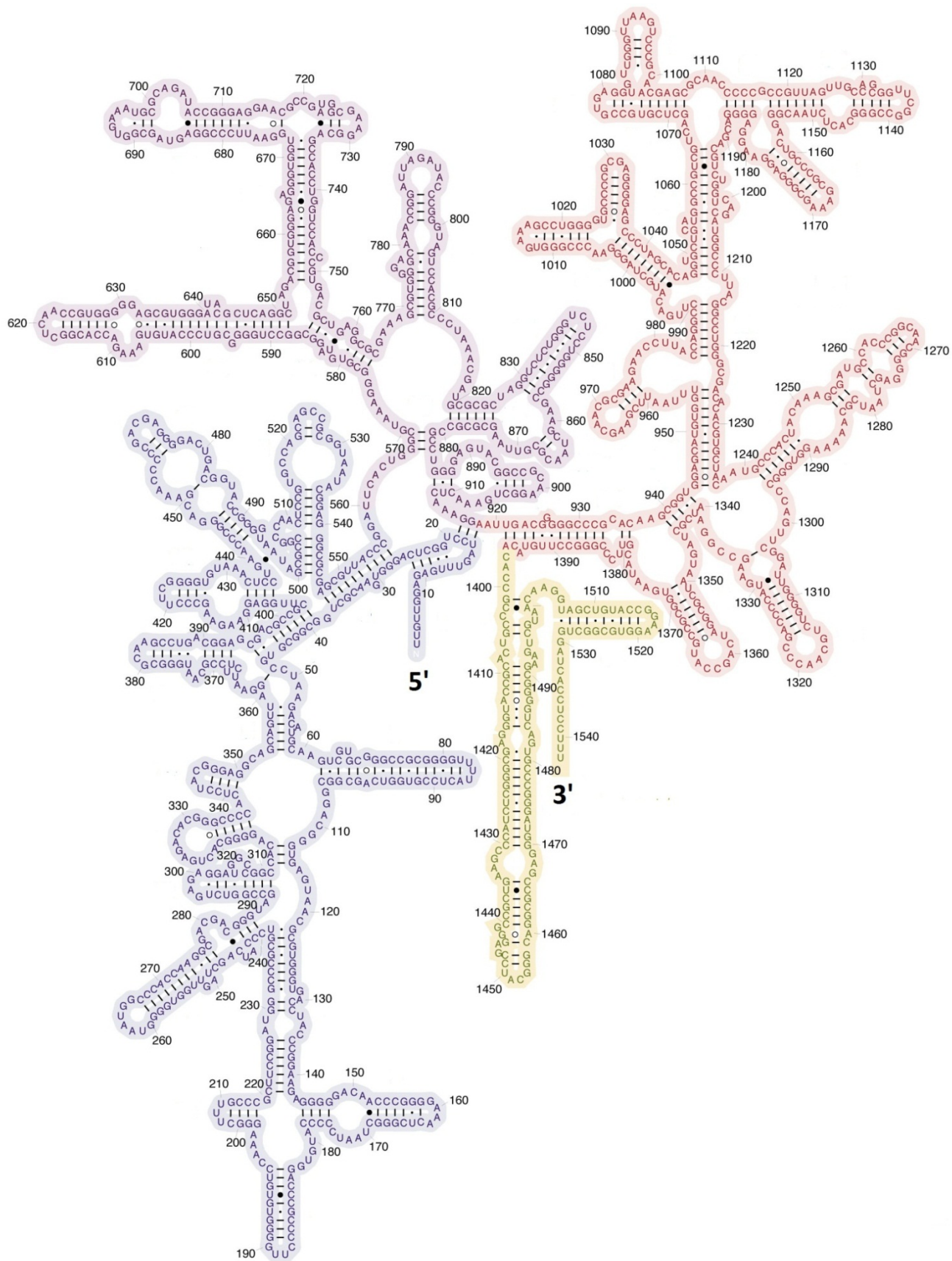


Fig. 2.2

- (c)** 16S rRNA and tRNA are both made up of four types of ribonucleotides linked by phosphodiester bonds.

State two other similarities between the structures of the 16S rRNA and tRNA. [2]

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- (d)** Describe how eukaryotic ribosomes are formed. [4]

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Unlike eukaryotes, prokaryotes do not have endoplasmic reticulum or vesicles.

- (e)** Suggest how prokaryotes are still able to secrete proteins. [1]

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[Total: 12]

QUESTION 3

The Human Immunodeficiency Virus (HIV) weakens the immune system, causing the patient to succumb to secondary infections, leading to the development of AIDS. The first recognized case of AIDS was reported in 1981 in Los Angeles, U.S.A.

Fig. 3.1 shows the binding of a HIV particle to a T cell. The viral receptor gp120 trimer binds to both CD4 and a co-receptor called CCR5 on the T cell surface.

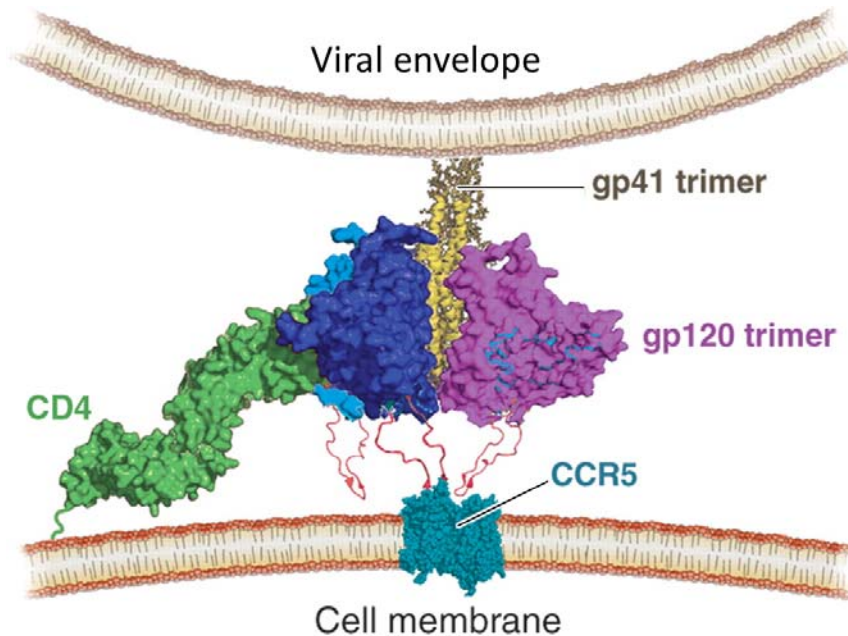


Fig. 3.1

- (a) Explain why the composition of the viral envelope is similar to the composition of the T cell membrane. [1]

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- (b) Individuals who carry a mutation known as CCR5- Δ 32 in both copies of the CCR5 gene are resistant to certain strains of HIV. The CCR5- Δ 32 allele is a result of a deletion of a particular sequence of 32 base-pairs within the exon of CCR5 gene.

Explain how an individual who is homozygous for the CCR5- Δ 32 allele is resistant to HIV. [4]

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- (c) It was observed that this strain of HIV that uses CCR5 as the co-receptor quickly evolves into another strain which bind to another co-receptor known as CXCR4.

Suggest and explain how HIV can evolve to use CXCR4 as the co-receptor. [2]

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- (d) The allele frequency of CCR5-Δ32 is unusually high in the western European countries, with up to 20% of ethnic western Europeans carrying this allele, which is rare or absent in all other ethnic groups throughout the world.

Suggest a reason for this phenomenon. [1]

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- (e) A technique called RNA interference has been shown to slow down the replication of HIV *in vitro*. Double-stranded RNA molecules, 21 to 23 nucleotides long, were added to a culture of T cells infected with HIV. The sequence of this small interfering RNA (siRNA) matches part of the HIV protease gene. Once inside the T cell, the two strands of the siRNA separate into single strands. One of the strands is identical in sequence to the mRNA of the HIV protease.

Explain how this siRNA is able to interfere with HIV replication. [4]

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[Total: 12]

QUESTION 4

Gene expression in eukaryotic cells can be controlled in many ways, one of which is the regulation of mRNA production. Fig. 4.1 shows how transcription can be controlled.

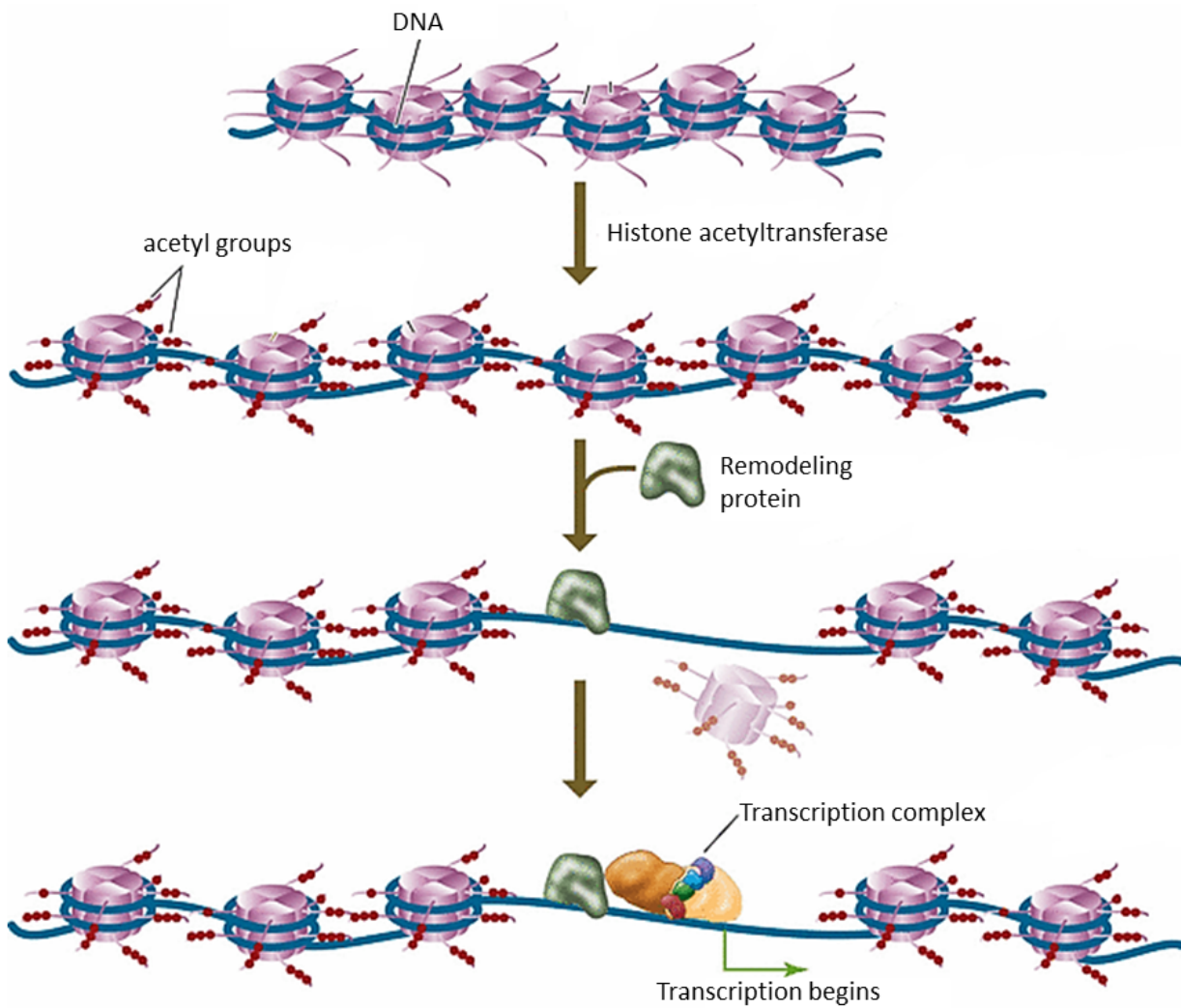


Fig. 4.1

- (a) Using the information in Fig. 4.1 and with your knowledge on the control of eukaryotic gene expression, explain how transcription is initiated. [4]

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Ubiquitin-dependent protein degradation in the proteasome is one way in which the amount of cytosolic proteins in a cell can be controlled at the post-translational level. An amino acid sequence of proline, glutamic acid, serine and threonine (PEST) in a protein are often recognized by enzymes and tagged with ubiquitin proteins. Globular proteins in their correct conformation are not tagged for degradation.

- (b)** Using the above information, suggest why denatured cytosolic proteins can be tagged with ubiquitin. [1]

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- (c)** Transmembrane proteins are usually not degraded by proteasome.

Suggest how transmembrane proteins on the cell surface are degraded. [2]

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Ovalbumin gene from chicken was isolated and made single-stranded, which was then mixed together with its mature mRNA. The results were observed under an electron microscope. Fig. 4.2 shows the electron micrograph and its corresponding diagrammatic representation, showing the binding of the mRNA to certain regions of the DNA.

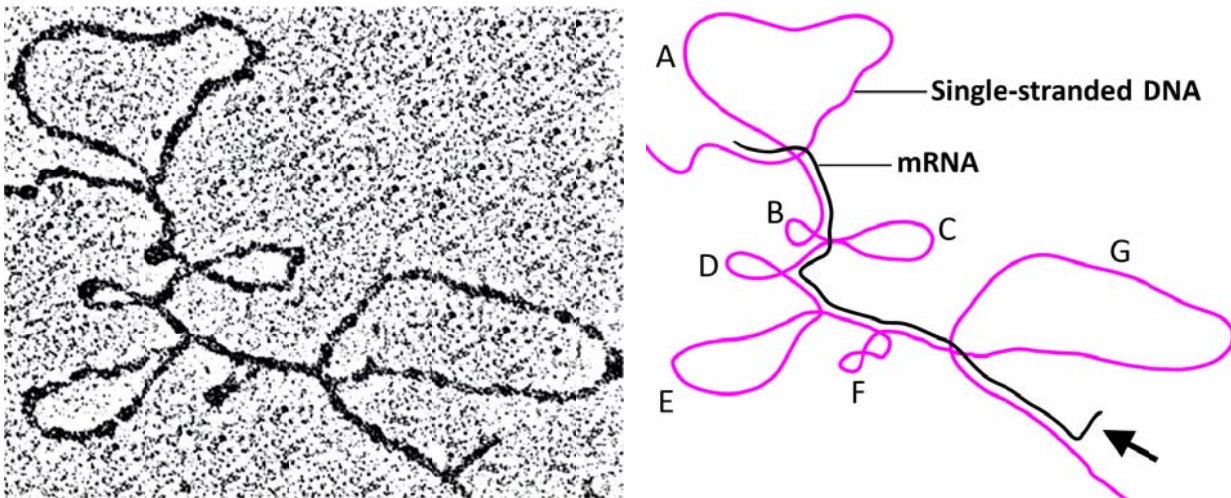


Fig. 4.2

(d) Explain why loops (A – G) were observed.

[3]

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(e) Explain why the region of the mRNA, indicated by the arrow in Fig. 4.2, remains unbound to DNA.

[2]

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Repeating nucleotide sequences are common in the genome of eukaryotes. These repeating sequences have been commonly referred to as 'junk DNA'.

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(f) Suggest why the term 'junk DNA' is misleading. [2]

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[Total: 14]

QUESTION 5

Hereditary hypophosphatemic rickets is a genetic disorder that results on low level of phosphate in the blood (hypophosphatemia).

Fig. 5.1 shows the inheritance of this disease over four generations in an extended family.

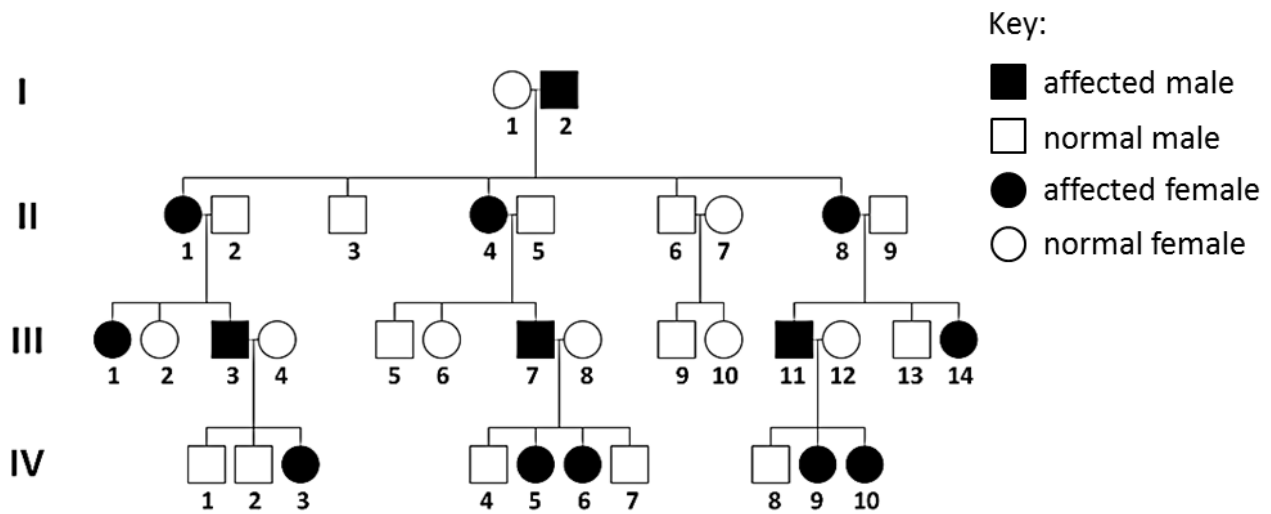


Fig. 5.1

(a) Explain the mode of inheritance of this disease.

[2]

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(b) Using the symbols **D/d** to represent the alleles, construct a genetic diagram to show how individual **III-7** and **III-8** can pass on the disease to their children.

[4]

Fur colour of Labrador dogs is controlled by two genes, **B/b** and **E/e**. The genes are on different pairs of chromosomes.

Allele **B** produces black fur while allele **b** produces brown fur. However, the colour is expressed only in the presence of allele **E**. In the absence of allele **E**, the dogs are beige in colour, hence the name Golden Labrador.

Crossing two heterozygotes typically gives the following phenotypic ratio:

9 black : 3 brown : 4 golden

- (c) Explain how the phenotypic ratio will differ if genes **B/b** and **E/e** are located on the same homologous pair of chromosomes, as shown in Fig. 5.2. [3]

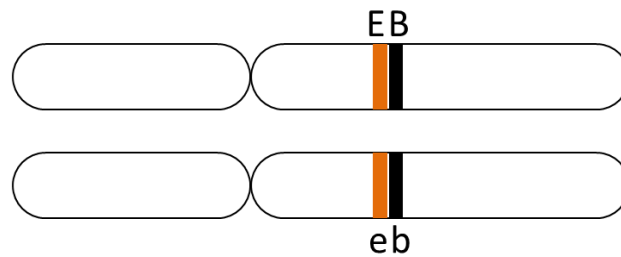


Fig. 5.2

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[Total: 9]

• End of Section A (Part I) •

Candidate Name

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Civics Group Index Number

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Meridian Junior College
JC2 Preliminary Examinations 2013
H2 Biology (9648/02)

Question 6	Question 7
/ 13	/ 10

Section A (Part II)
Answer **all** the questions in this section.

QUESTION 6

(a) Describe the role of the following ions in impulse transmission along a nerve cell.

(i) Na^+ [2]

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(ii) K^+ [2]

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(b) Describe the role of Ca^{2+} in the passage of impulses across a synapse. [3]

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(c) Explain why the concentration of Ca^{2+} in the cytosol of the synaptic knob remains low despite frequent opening of channels. [2]

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A water-soluble toxin, muscarine, derived from a mushroom, causes deceleration of the heart rate, resulting in death of individuals. Fig. 6.1 shows the signalling pathway of muscarine on a postsynaptic membrane.

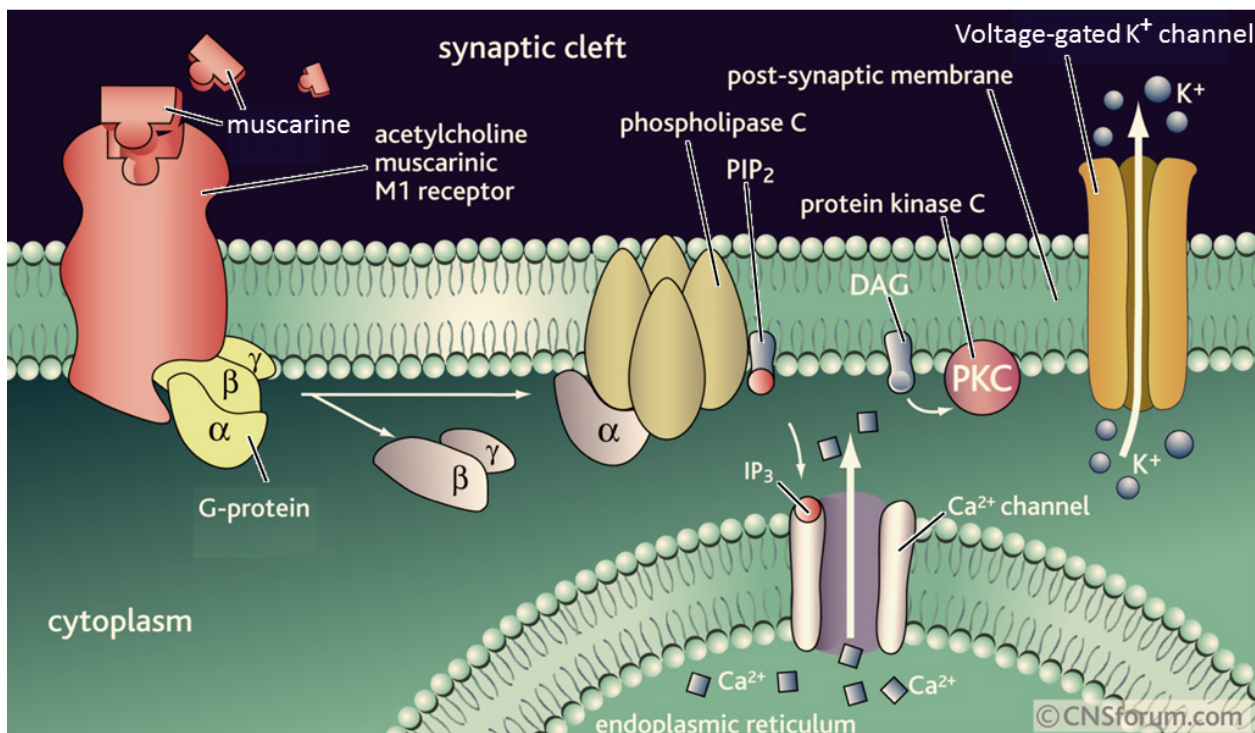


Fig. 6.1

(d) Describe how muscarine results in inhibition of nerve transmission in heart muscle cells. [4]

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[Total: 13]

QUESTION 7

Fig. 7.1 shows the distribution of malaria and the allele frequency of the sickle cell allele.

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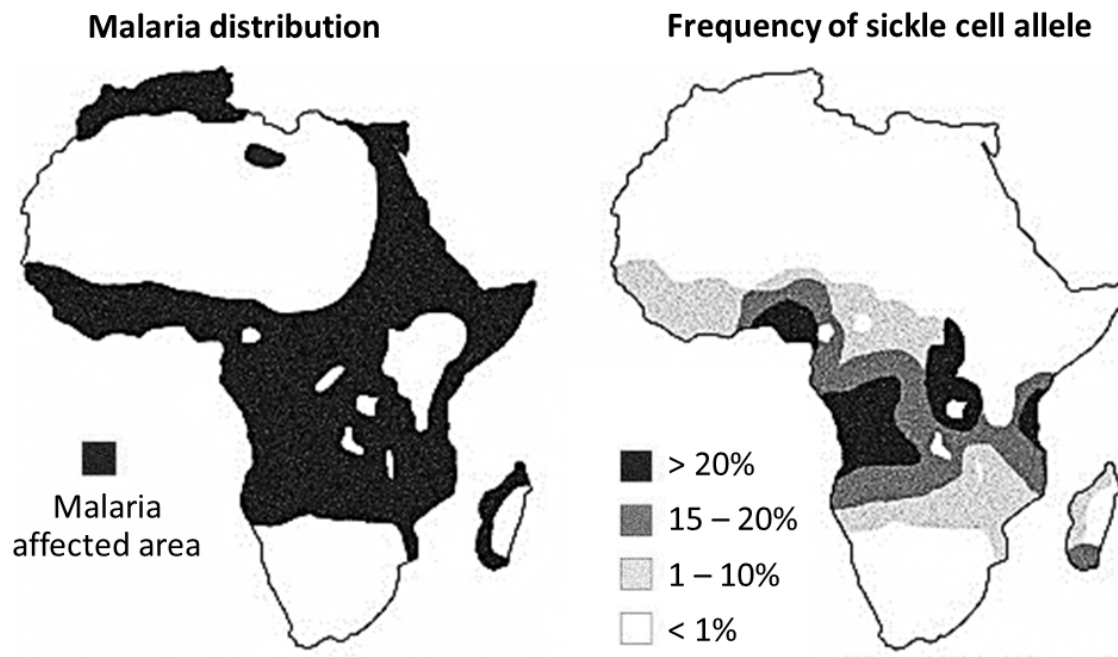


Fig. 7.1

- (a) Describe and explain the correlation between the occurrence of malaria and the frequency of the sickle cell allele. [4]

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The evolution of haemoglobin molecules has been studied extensively by comparing the amino acid sequences in both myoglobin and haemoglobin. Myoglobin is used for oxygen storage while haemoglobin is used for oxygen transport. Ancient prehistoric animals had a single chain of simple globin for oxygen storage and transport.

About 500 million years ago, a gene duplication event occurred and one copy became the present day myoglobin and the other evolved into an oxygen-transport protein that gave rise to the present day haemoglobin (Fig. 7.2).

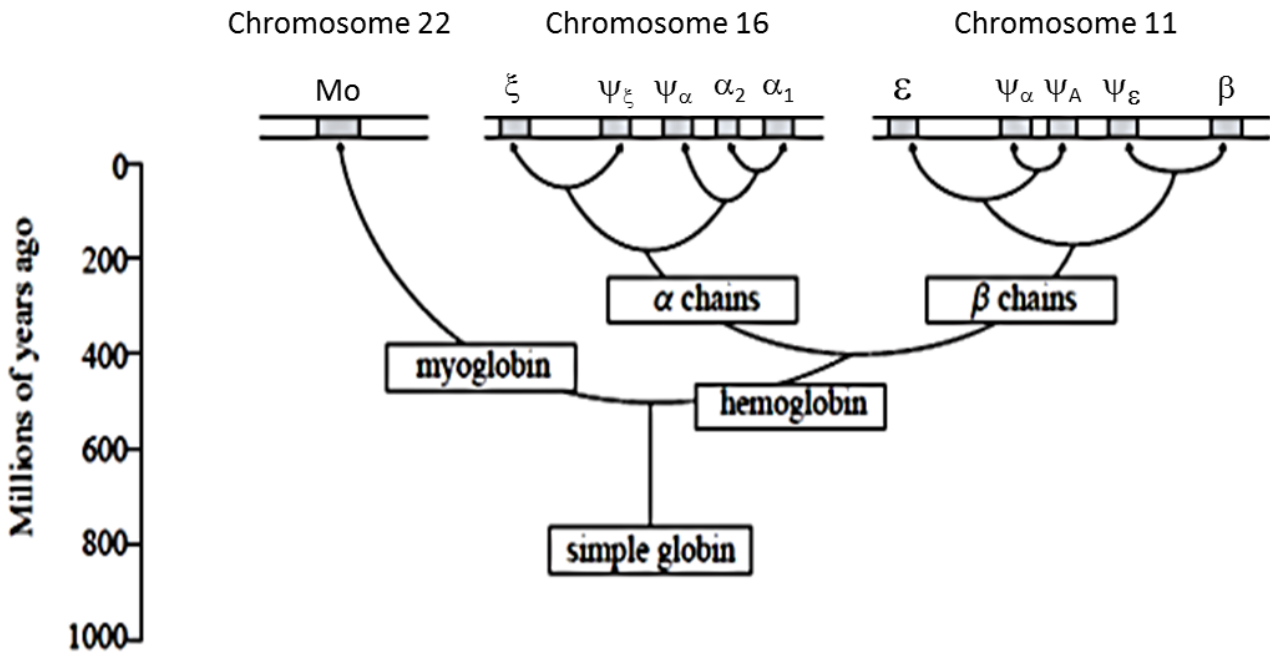


Fig. 7.2

(b) State how many years ago the haemoglobin molecule diverged into α chains and β chains. [1]

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(c) Suggest why changes observed in the sequence of amino acids may lead to an under-estimation of the actual number of mutations. [2]

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(d) Explain how variations in the haemoglobin gene can be used as a molecular clock. [3]

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[Total: 10]

• End of Section A (Part II) •

Section B
Answer **ONE** question

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Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in questions **(a)**, **(b)**, etc., as indicated in the question.

QUESTION 8

- (a)** Describe the importance of ATP in cellular processes. [8]
- (b)** Distinguish between a specialized cell and a cancerous cell. [5]
- (c)** Discuss, with a named example, how mutation to centromeres can result in genetic disorders. [7]

[Total: 20]

QUESTION 9

- (a)** Describe and explain how the structures of chloroplasts and mitochondria are similarly adapted for their functions. [8]
- (b)** Discuss the factors that contribute to speciation. [7]
- (c)** Explain the concept of negative feedback in controlling insulin levels in the body. [5]

[Total: 20]

• End of Section B •